



Real-time detection of flame and smoke using an improved YOLOv4 network

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Received: 24 July 2021 / Revised: 17 September 2021 / Accepted: 18 October 2021 / Published online: 17 January 2022
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Abstract

Fire is one of the major disasters in the world, which seriously endangers the safety of life and property. Effective flame and smoke detection can provide timely warning information for firefighters. Existing flame and smoke detection algorithms are limited by processor performance and cannot operate large deep networks. This paper proposes a lightweight detector called Light-YOLOv4, which considers the balance between performance and efficiency. First, the backbone network CSPDarknet53 is replaced by MobileNetv3. Second, the path aggregation network is changed into bidirectional feature pyramid network (BiFPN) based on the idea of bidirectional cross-scale connections. Third, an efficient feature extraction module named depthwise separable attention module based on depthwise separable convolution and coordinate attention network is construed to replace the 3×3 standard convolution of spatial pyramid pooling, BiFPN and YOLO head network. In comparison with YOLOv4, Light-YOLOv4 has only 19.1% of its trainable parameters while almost keeping the same accuracy. By combining multiple tricks, Light-YOLOv4 can achieve a better balance between performance (85.64% mAP) and efficiency (71 FPS), which meets flame and smoke detection tasks' requirements on the accuracy and real time. Experiments on Nvidia Jetson TX2 further demonstrate that Light-YOLOv4 has good detection performance and speed on embedded scenarios.

Keywords Flame and smoke detection · YOLOv4 · MobileNetv3 · DSAM · BiFPN · Light-YOLOv4

1 Introduction

Fire is one of the major disasters threatening public safety, causing a large number of casualties and economic losses every year. The early detection of fire is extremely more important than anything else to minimize loss of lives and property damage [1].

Flame and smoke detection usually includes methods based on traditional sensors and methods based on image sensors. The methods based on traditional machine learning methods and deep learning methods. Traditional machine learning methods are less dependent on data and processor performance, but their performance depends on the accuracy of manually extracted features which are color [2], shape, texture [3], and motion [4]. Recent advances in artificial

intelligence and machine learning have made image-based modeling and analysis even more successful in different applications [5]. Also, with the advent of nanotechnology semiconductors, a new generation of tensor processing units (TPUs) and graphical processing units (GPUs) can provide an extraordinary computation capability for data-driven methods [6]. For these reasons, the fire detection method based on convolutional neural network has begun to attract the attention of researchers.

The purpose of this paper is to solve the problem that traditional deep learning methods are not suitable for low-end processors. This paper improves the YOLOv4 [7] method from three aspects. On the one hand, the lightweight backbone network is used to replace the backbone network of YOLOv4. Then, we take advantage of bidirectional cross-scale connections to guide YOLOv4. BiFPN [8] can shorten the path from low-level information to high-level information, and construct the residual structure of feature pyramid network, so as to integrate richer semantic features and retain spatial information. On the other hand, we decompose the standard convolution, calculate the channel and spatial region separately and traverse each channel, obtaining the mapping

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equivalent to the number of channels. In this way, the number of network parameters can be greatly reduced, and the network lightweight can be further realized.

2 Related work

The traditional method is based on artificially designed features. Through the collection of color, texture, shape, and other features, the BP neural network or support vector machine (SVM) is used for training, and the training model is used to detect the video frame [9]. Chino et al. [10] proposed BoWFire, which reduces the number of parameters and the possibility of false-positive, and simplifies the fine-tuning process. Wang et al. [11] proposed a flame detection method based on KNN background subtraction that combines the flame color feature and the local feature. Osman et al. [12] propose a real-time dynamic texture recognition method using projections onto random hyperplanes and deep neural network filters and propose an application of the proposed method to real-time detection of flames in infrared videos.

With the rapid development of object detection methods based on deep learning, the latest object detection methods have been applied to the field of flame and smoke detection. Yu et al. [13] proposed a flame detection algorithm based on two-stream convolution neural network combining spatial and temporal features. ChaoXia et al. [14] proposed an improved Faster-RCNN flame detection method. This method uses the color features of the flame to define the limit of the anchor and uses a parallel network to generate image global information to guide the detection. Hongyi Pan et al. [15] propose a deep convolutional neural network for camera-based wildfire detection which uses window-based analysis strategy to increase the fire detection rate. Hongyi Pan et al. [16] also propose a video-based wildfire detector called AddNet. They constructed a dot product-like operation from the mf-operator and use it to define dense and convolutional feed-forwarding passes.

3 Methods

This paper presents a methodology called Light-YOLOv4. The backbone network is MobileNetv3, which is used for preliminary feature extraction. The detection neck network is composed of spatial pyramid pooling (SPP) [17] and BiFPN for further feature extraction. The SPP module is located after MobileNetv3. After performing three convolution operations on the last feature layer of MobileNetv3, use 13×13 , 9×9 , 5×5 , and 1×1 pooling kernels for max-pooling operations. BiFPN enhances the process of instance segmentation by preserving spatial information, which helps to locate pixels

correctly. Compared with feature pyramid networks (FPN) [18], path aggregation network (PAN) adds a bottom-up path, and BiFPN further adds a top-down path, thereby shortening the information path between low-level and high-level features. The three input feature layers of BiFPN come from MobileNetv3 and SPP, and their shapes are $52 \times 52 \times 40$, $26 \times 26 \times 112$, $13 \times 13 \times 512$, respectively. The network structure of Light-YOLOv4 is shown in Fig. 1.

3.1 MobileNetv3

MobileNet [19–21] is a lightweight convolutional neural network focused on mobile devices and embedded devices. Compared with the traditional convolutional neural network, MobileNet greatly reduces the model parameters and the amount of calculation under the premise of only a little reduction in accuracy.

In addition to the use of depthwise separable convolution and reverse residual structure, MobileNetv3 introduces the squeeze-and-excitation (SE) [22] channel attention network in the reverse residual structure, which can effectively construct the weight relationship between channels and improve network efficiency.

3.2 BiFPN

YOLOv4 uses PAN on the neck for feature extraction and fusion. Although it solves the problem of large differences in object scales in different scene images, it is not for the optimal solution of feature fusion, EfficientDet proposes a better feature fusion method BiFPN.

Light-YOLOv4 combines the idea of bidirectional cross-scale connections, it connects input nodes and output nodes at the same level across layers, shortening the path of low-level information transmission to high-level, and combining the rich semantic features of high-level with those located at a lower level. We merge adjacent layers by concatenating instead of adding to improve the accuracy of prediction. We also removed the weighted feature fusion of BiFPN, so as not to reduce the inference speed of the network.

3.3 Depthwise separable attention module

Depthwise separable convolution resolves standard convolution into depthwise convolution and pointwise convolution. Depthwise convolution is to obtain the spatial information of the feature, and then perform pointwise convolution on the output feature, that is, use a 1×1 convolution kernel to convolve the output of the depthwise convolution. The process of depthwise convolution and pointwise convolution is shown in Fig. 2. This combination can achieve a lightweight effect.

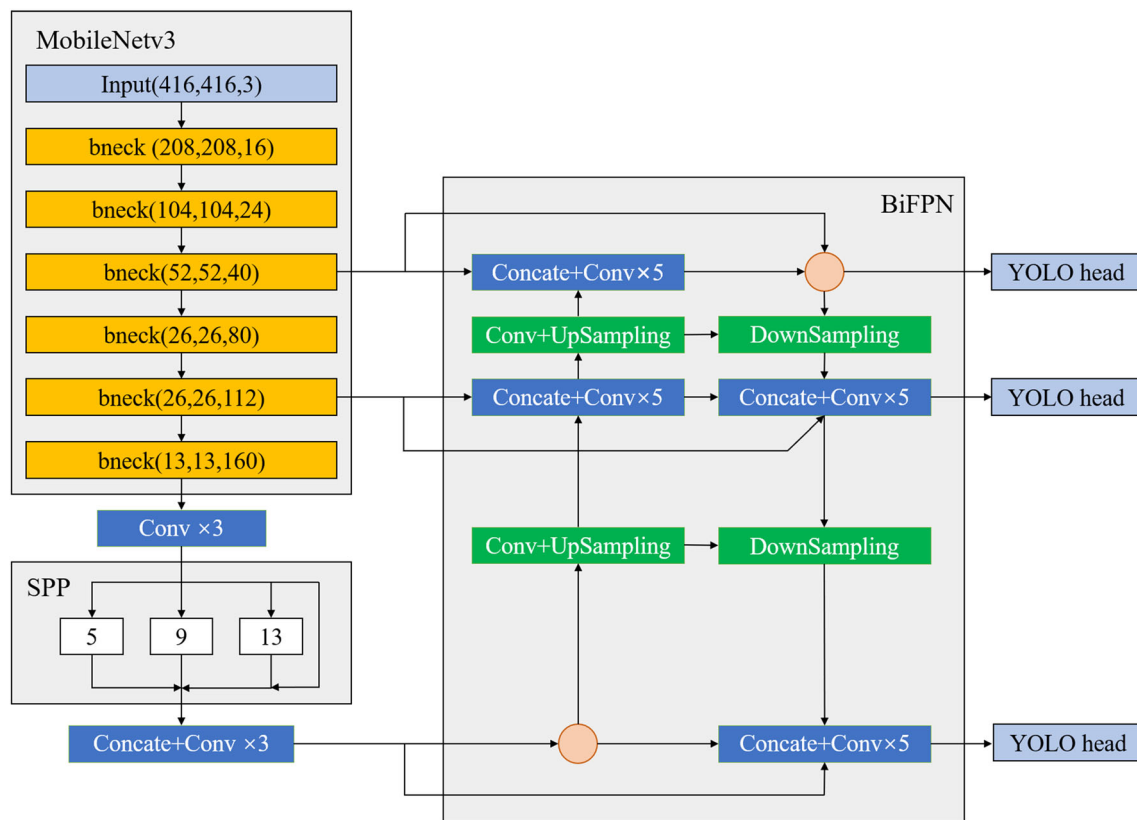


Fig. 1 The network architecture of Light-YOLOv4

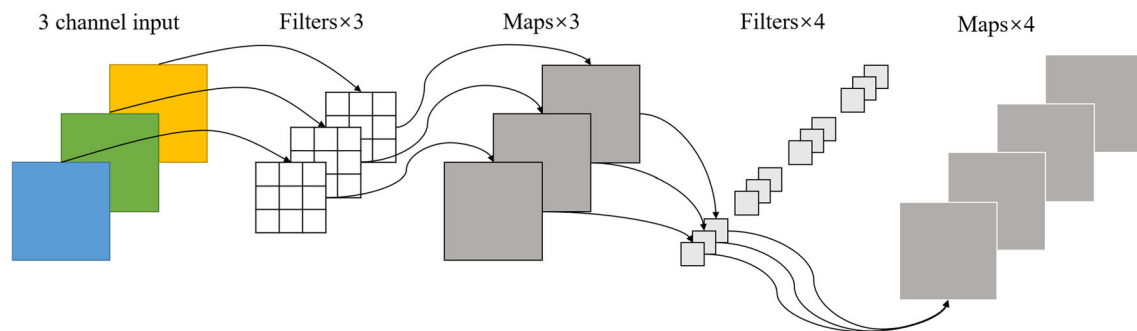


Fig. 2 Depthwise convolution

Channel attention can improve the performance of neural network by encoding the dependencies between channels, but they ignore the location information, which is very important for generating the spatial structure. coordinate attention network (CAN) [23] solves the above problems by embedding position information into the channel attention.

CAN converts the original channel attention network and divides it into two encoding processes. One encoding process is used to capture the dependency between channels, and the other encoding process is used to capture position information. Then, the two obtained feature maps are coded

into a pair of direction-aware and position-sensitive attention maps, which are applied to the input feature maps to obtain the object of interest. Based on the above research, combined with depthwise separable convolution, a depth separable attention module DSAM is proposed.

After the depthwise convolution, it is followed by global average pooling, and it is decomposed into two one-dimensional feature encoding operations through two pooling kernels. One channel uses a pooling kernel with a size of $(H, 1)$ to encode the channel along the horizontal coordinate. Given an input X , the output of the c -th channel with height

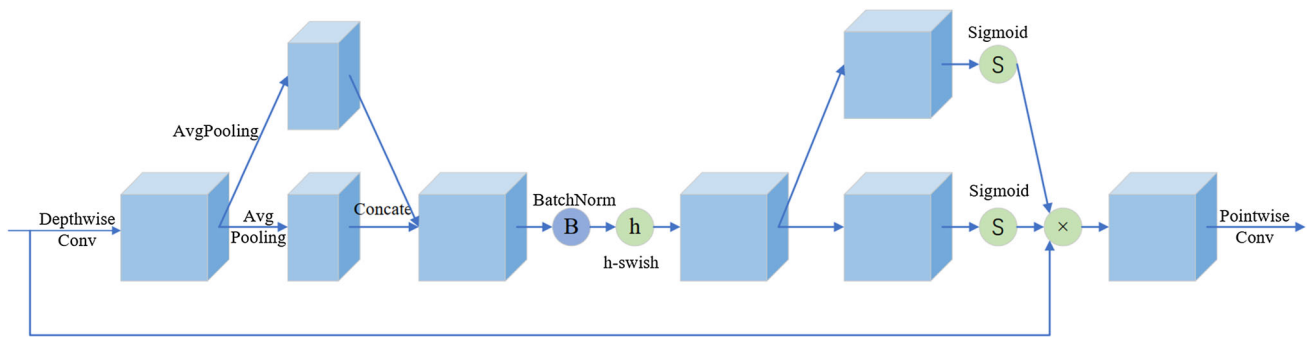


Fig. 3 The network architecture of depthwise separable attention module (DSAM)

h Eq. (1).

$$z_c(h) = \frac{1}{W} \sum_{0 \leq i < W} x_c(h, i). \quad (1)$$

Another channel uses a pooling kernel with a size of $(1, W)$ to encode the channel along the vertical coordinate. The output of the c -th channel with width w can be expressed by Eq. (2).

$$z_c(w) = \frac{1}{H} \sum_{0 \leq j < H} x_c(j, w). \quad (2)$$

Features are aggregated along the height and width directions, respectively, through the above two pooling operations, and the position information in the global spatial information is obtained, which is beneficial to the network to better locate the object of interest. Then, perform concatenate splicing operation on the feature maps obtained by the above two channels, use 1×1 convolution to convert the number of channels, and then use batch normalization and h-swish activation function to obtain global spatial information in the horizontal and vertical directions. The output of intermediate feature map can be expressed by Eq. (3).

$$f = \delta(F_1([z^h, z^w])). \quad (3)$$

In Eq.(3), $[z^h, z^w]$ represents the splicing operation along the horizontal and vertical directions. δ is h-swish activation function and f is the intermediate feature map that encodes through horizontal and vertical direction.

Then, divide the intermediate feature map into two separate tensors along the spatial dimension, and use two 1×1 convolutions to convert the channels to be consistent with the input channels. The conversion process can be expressed by Eqs. (4) and (5).

$$g^h = \sigma(F_h(f^h)) \quad (4)$$

$$g^w = \sigma(F_w(f^w)). \quad (5)$$

In Eqs. (4) and (5), F_w and F_h represent the 1×1 convolutional transformations and σ represents the sigmoid activate function. The two tensors g^h and g^w obtained are used as attention weights. In conclusion, the output of coordinate attention network can be expressed by Eq. (6). The network structure diagram of DSAM is shown in Fig. 3.

$$y_c(i, j) = x_c(i, j) \times g^h(i) \times g^w(j). \quad (6)$$

4 Experiment

4.1 Datasets

This dataset motivates researchers to seek novel solutions for early fire detection and management. In order to verify the rationality of Light-YOLOv4 in flame and smoke detection tasks, we collected flame and smoke pictures to construct a flame and smoke dataset. This dataset contains almost 40,000 images, which contains two types of objects, the labels are fire and smoke, respectively. Our dataset contains various scenarios, simple backgrounds to complex backgrounds, single objects to multiple objects, indoor fires into forest fires, small fires to large fires, ensuring the generalization of the model. Part of the dataset image is shown in Fig. 4.

4.2 Training details

This section mainly introduces the dataset and training details used in the experiment. The hardware platforms used in training is NVIDIA GeForce RTX 2070 SUPER with 8 GB video memory. In our experiment, 19,189 images were divided into training dataset, development dataset, and test dataset according to the ratio of 7:1:2. Before training, clustering is performed by K-means algorithm, and the anchor box is initialized.



Fig. 4 Part of the dataset image

Table 1 Ablation study on the flame and smoke dataset

	MobileNetv3	DSAM	BiFPN	Params(M)	mAP(%)	mAP@0.5(%)	FPS
YOLOv4	–	–	–	243.92	86.08	71.08	40
	✓	–	–	152.55	85.38	69.32	58
	–	✓	–	137.18	85.65	70.49	56
	–	–	✓	249.55	86.34	71.26	39
	✓	✓	–	45.44	85.58	69.88	72
	✓	✓	✓	46.69	85.64	70.88	71

4.3 Ablation experiment on training platform

In order to verify the impact of various improvements of Light-YOLOv4 on the performance of the detector, we performed a series of additional tests for three methods on training platform, as shown in Table 1.

In conventional object detection tasks, we often use mAP (mean average precision), mAP@0.5, precision and recall as the criterion for measuring the model performance. Generally speaking, average precision is the area enclosed by the precision–recall curve. And mAP is obtained by comprehensively weighted average of average precision detected in all categories, mAP@0.5 represents the evaluation index when IoU (Intersection over Union) is equal to 0.5. IoU refers to the ratio of the intersection area of ground truth bounding box and predict bounding box to their union area. The precision and recall can be defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (7)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (8)$$

Among them, TP is the number of positive samples predicted as positive samples. FP is the number of negative samples predicted as positive samples. FN is the number of positive samples predicted as negative samples. TN is the number of negative samples predicted as negative samples.

After the introduction of DSAM, the detection speed is further improved, and the accuracy loss caused by the lightweight of the network structure is supplemented to a

certain extent. On the other hand, one can see that BiFPN can effectively improve the feature fusion capability of the network. YOLOv4 with BiFPN achieves the highest detection accuracy. At the same time, the use of BiFPN has little impact on the detection speed, and the negative influence is further reduced through the combined use with DSAM.

Light-YOLOv4 achieves a good balance of performance and efficiency. The parameters of the network are 46.69M, which only accounts for 19.1% of YOLOv4. The mAP and mAP@0.5 reach 85.64% and 70.88%, respectively, which are only 0.44% and 0.20% lower than YOLOv4, but the inferencing speed is 77.5% faster than YOLOv4, reaching 71FPS. The precision of Light-YOLOv4 for flame and smoke reached 95.91% and 97.90%, respectively, and the recall reached 69.10% and 64.68%, respectively, which shows the high detection precision and low false detection rate. The precision–recall curve of Light-YOLOv4 is shown in Fig. 5.

4.4 Experimental results and discussion

We verify our Light-YOLOv4 on the test images. It can be seen from Figs. 6 and 7 that Light-YOLOv4 has a good detection effect on large and medium-sized fires and early fires, but the detection effect of overlapping objects is not as good as single objects. This is because Light-YOLOv4 divides the image into $n \times n$ grid cells, and each grid cell is only responsible for detecting one object, which leads to the omission of overlapping objects. If we want to improve the detection effect of overlapping objects, we can divide the image into more grid cells, but this will increase the parameters.

Fig. 5 Precision–recall curve of Light-YOLOv4

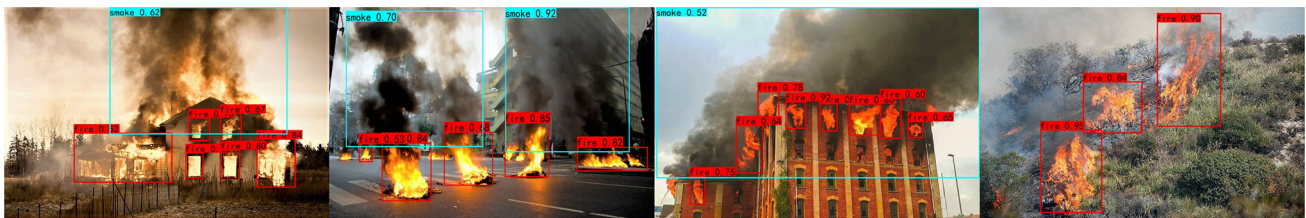
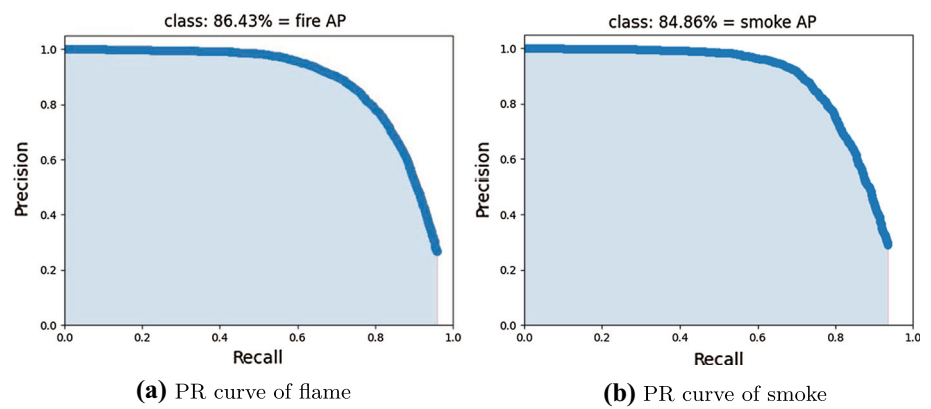


Fig. 6 Detection results of Light-YOLOv4



Fig. 7 Detection results of Light-YOLOv4 for early fire

4.5 Compared with other methods

In order to further verify the specific performance of our flame and smoke detection methods, we have compared and verified with the commonly used methods in the literature in recent years. The performance test results of the above methods are shown in Table 2. These results show that the combination of MobileNetv3, DSAM, and BiFPN got the best balance in speed and precision. It is only slower than YOLOv4-tiny in speed and only lower than YOLOv4 in precision.

We also verified the detection rate and false alarm rate of this work. It is hard to compare with other methods because most of them are not open source. Here we use a common dataset BoWFire for comparison. It contains 119 fire images

and 107 no-fire images, and it has been used as the test dataset by many related works [19]. The comparison results are shown in Table 3. We can see that the false alarm rate of Light-YOLOv4 is slightly higher, but the detection rate and accuracy are better.

4.6 Deployment experiment of AMOVLAB P600 UAV

We deploy the trained Light-YOLOv4 model to the AMOVLAB P600 UAV shown in Fig. 8. We called the onboard camera on the embedded GPU TX2 platform for forward inference, and read the video from the mobile phone for real-time display. It can be seen that the real-time detection frame number can reach 10 FPS, and there is no obvious error detection and misjudgment phenomenon. It meets real-

Table 2 Comparison of different methods

Method	Params(M)	mAP(%)	mAP@0.5(%)	FPS
Faster-RCNN [24]	107.98	84.35	67.02	12
SSD [25]	90.58	85.29	68.87	50
YOLOv3 [26]	234.71	84.52	70.30	43
YOLOv4	243.92	86.08	71.08	40
YOLOv4-tiny	22.58	77.61	62.63	140
EfficientDet-D2	31.2	80.61	66.43	17
Light-YOLOv4	46.69	85.64	70.88	71

Table 3 Comparison of different methods

Method	Detection rate (%)	False alarm rate (%)	Accuracy (%)
Muhammad et al. [27]	97.48	18.69	89.82
Muhammad et al. [28]	93.28	9.34	92.04
Chaoxia et al. [14]	92.44	5.61	93.36
Hongyi et al. [20]	91.60	4.67	93.36
Light-YOLOv4	93.14	6.35	95.24

**(a)** AMOVLAB P600 UAV**(b)** Result**(c)** Result**Fig. 8** AMOVLAB P600 UAV and detection result

time and accuracy requirements and can be used for actual flame and smoke detection tasks.

5 Conclusion

This paper proposes an improved YOLOv4 lightweight object detector Light-YOLOv4. The lightweight detection network MobileNetv3 is used as the backbone network to replace CSPDarknet, which reduces the network parameters while maintaining high accuracy. Based on the depthwise separable module and coordinate attention, the depthwise separable attention module DSAM is proposed to further reduce the parameters of the network, and through the attention mechanism to increase the weight of the network to the region of interest and improve the detection Accuracy. The BiFPN feature fusion network improves the efficiency of multi-scale feature fusion and improves the detection accuracy of the network by introducing top-down pathway, bottom-up pathway, and cross-scale connection. Construct flame and smoke dataset, use K-means algorithm for clustering, and initialize anchor boxes. Experimental results show

that the Light-YOLOv4 detector has a flame detection accuracy of 86.43%, a smoke detection accuracy of 84.86%, mAP of 85.64%, mAP%0.5 of 70.88%, and FPS of 71 in the tasks of flame and smoke detection. Compared with the YOLOv4 detector, when mAP is only reduced by 0.44%, the parameters are reduced by 80.9%, and the FPS is increased by 77.5%. The detection precision and detection speed of Light-YOLOv4 are more balanced, which is suitable for flame and smoke detection tasks on embedded hardware platforms. Compared with the existing flame and smoke detection methods based on deep learning, Light-YOLOv4 is also advanced.

Acknowledgements This work was supported by the National Key R & D Program of China (2019YFB1312104) and Key R & D Program of Hebei Province (20311803D).

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